Math Kangaroo Practice

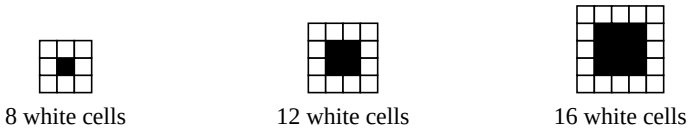
Grade 3-4 · 24 questions · standard · seed 20261020

Part A: Each question is worth 3 points

1. How many digits are needed to write down all the numbers from 1 to 100?

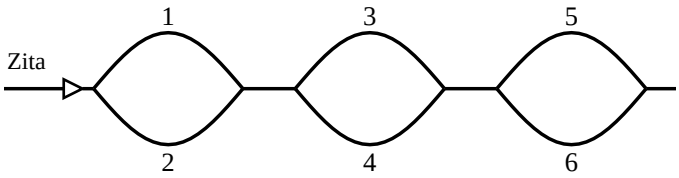
A. 210 ***B.*** 200 ***C.*** 193 ***D.*** 192 ***E.*** 189

2. These diagrams follow a pattern: a black square with a white border around it. How many white cells will the next diagram have?



A. 36 ***B.*** 24 ***C.*** 23 ***D.*** 20 ***E.*** 16

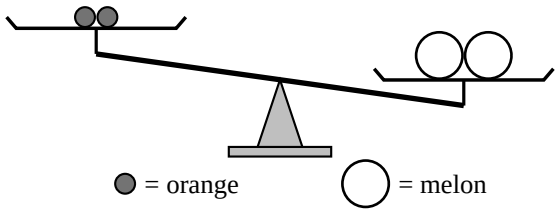
3. Zita walks along the road from left to right and writes down every number on the way.



Which group of numbers could Zita have written?

A. 3, 4 and 5 ***B.*** 4, 5 and 6 ***C.*** 1, 4 and 5 ***D.*** 1, 3 and 4 ***E.*** 1, 2 and 5

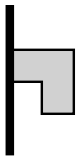
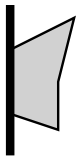
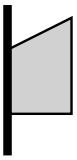
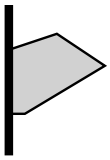
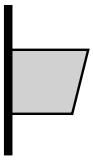
4. On one plate of a balance there are 2 oranges and on the other there are 2 melons. When we put one more melon, exactly like the others, on the plate with the oranges, the balance is level.



One melon weighs the same as how many oranges?

A. 1 orange ***B.*** 2 oranges ***C.*** 3 oranges ***D.*** 4 oranges ***E.*** 5 oranges

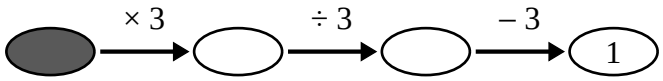
5. A rectangular flag flies on a pole in front of a wall, and the sun throws its shadow onto the wall. The wind turns and folds the flag, but the side sewn to the pole always stays on the pole. Leo draws the shadow at five different moments.
Which picture cannot be right unless the flag is torn?

A.  ***B.***  ***C.***  ***D.***  ***E.*** 

6. Sofia, Noah, Julien and Leo each play a different instrument: the guitar, the violin, the piano and the flute. Noah blows into the instrument. Sofia presses keys. Leo does not need a bow.
Who plays the guitar?

A. Sofia ***B.*** Noah ***C.*** Julien
D. Leo ***E.*** Impossible to determine

7. Tom wrote a favourite number in the dark cloud and did the calculations shown by the arrows, one after the other. The result was 1.



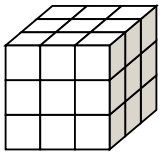
What is Tom's favourite number?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 4

8. Jumpy the kangaroo's longest jump in training was 30 dm 30 cm and 50 mm long. At the competition the jump was 117 cm longer.
How long was the jump at the competition?

A. 1 m 82 cm ***B.*** 3 m 35 cm ***C.*** 4 m 52 cm ***D.*** 4 m 97 cm ***E.*** 15 m 5 cm

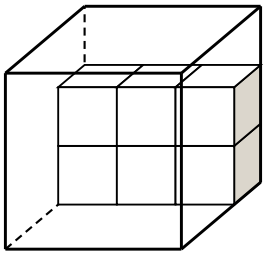
9. A cube with sides of 3 cm is painted grey all over and then cut into small cubes with sides of 1 cm.



How many of the small cubes have exactly 2 painted faces?

A. 12 ***B.*** 11 ***C.*** 8 ***D.*** 6 ***E.*** 4

10. Sara has a glass aquarium shaped like a cube, 3 dm long, 3 dm wide and 3 dm tall. Sara has started to put cubes with edges of 1 dm into it, as in the picture. Every cube stands on the bottom or on another cube.



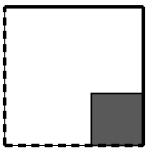
At most how many more such cubes can Sara put into the aquarium?

A. 21 ***B.*** 20 ***C.*** 18 ***D.*** 16 ***E.*** 6

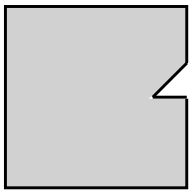
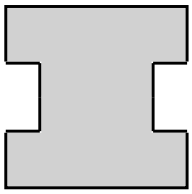
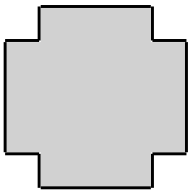
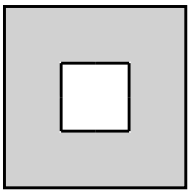
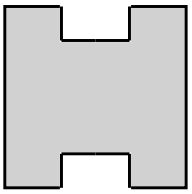
11. The digits from 1 to 9 are written on 9 cards. Lily has 1, 3 and 9; Sara has 2, 4 and 5; Martha has 6, 7 and 8. Each of them uses some of +, –, ×, ÷ (and brackets) and each of their own cards exactly once.
Who *cannot* get 14 as the result?

A. Lily ***B.*** Sara ***C.*** Martha
D. All can get 14 ***E.*** Sara and Martha

12. A square piece of paper is folded in half, and in half again, so that it is a small square (the dashed sides are the folds). The dark part is cut off, then the paper is unfolded.



What does the paper look like?

A.  ***B.***  ***C.***  ***D.***  ***E.*** 

Part B: Each question is worth 4 points

13. Omar was born on February 24, 2008 and Leo on December 16, 2009.
How many years old is Leo on June 15, 2016?

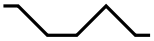
A. 8 ***B.*** 7 ***C.*** 6 ***D.*** 5 ***E.*** 4

14. Tom builds a brick 4 cubes long and 4 cubes tall from red and blue cubes. The outside is completely red and the inside is completely blue. Tom used exactly 8 blue cubes.
How many cubes deep is the brick?

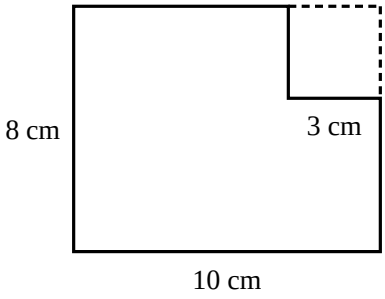
A. 2 ***B.*** 4 ***C.*** 5 ***D.*** 6 ***E.*** 7

15. Far away we see the skyline of a castle.

Which of the pieces **can** belong to the skyline?

A.  ***B.*** ***C.*** ***D.*** ***E.***

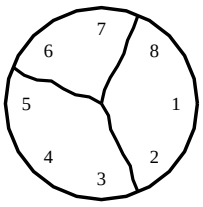
16. A rectangle is 10 cm long and 8 cm wide. A square with side 3 cm is cut off one corner.

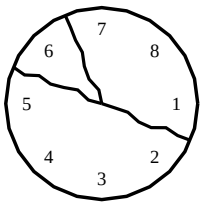


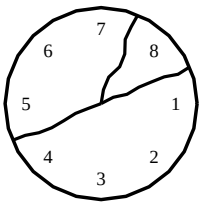
What is the perimeter of the remaining shape?

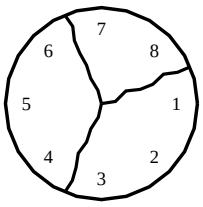
A. 36 cm ***B.*** 33 cm ***C.*** 30 cm ***D.*** 24 cm ***E.*** 18 cm

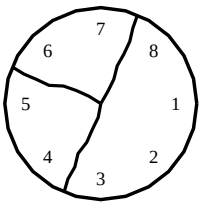
17. A round cake with the numbers 1 to 8 around its edge is cracked into 3 pieces. The sums of the numbers in the pieces are three consecutive numbers.
Which picture shows how it is cracked?

A. 

B. 

C. 

D. 

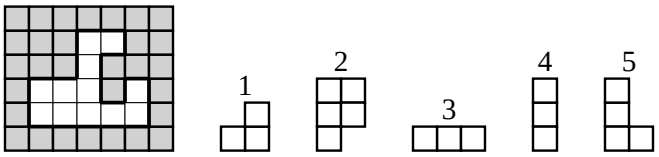
E. 

18. Leo reads a chapter from page 14 to page 25 (both included).
How many digits are printed in the page numbers of this chapter?

A. 12 ***B.*** 18 ***C.*** 22 ***D.*** 24 ***E.*** 41

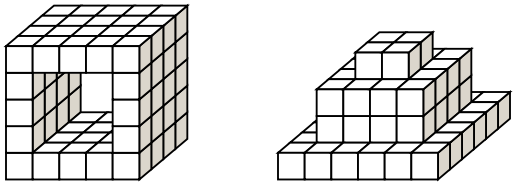
Part C: Each question is worth 5 points

19. Which three of these pieces can be used to cover the empty area exactly? The pieces may be turned and flipped over.



A. 1+2+3 ***B.*** 1+2+5 ***C.*** 1+3+5 ***D.*** 2+3+4 ***E.*** 2+4+5

20. Nina has the tunnel on the left. Nina takes it apart to build the pyramid on the right, but there are not enough cubes. Nothing is hollow inside the pyramid.

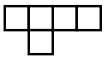


tunnel pyramid

How many more cubes does Nina need?

A. 8 ***B.*** 9 ***C.*** 10 ***D.*** 24 ***E.*** 56

21. Mia has two copies of the same piece. With which piece can she make a rectangle? Pieces may be turned and flipped over.

A.  ***B.*** ***C.*** ***D.*** ***E.***

22. Four children, Emma, Ben, Grace and Clara, sit in a row of 4 chairs at the cinema. Emma and Ben are best friends and want to sit next to each other.
In how many different ways can the children sit?

A. 24 ***B.*** 15 ***C.*** 14 ***D.*** 12 ***E.*** 3

23. How many two-digit numbers have digits that add up to 8?

A. 16 ***B.*** 8 ***C.*** 7 ***D.*** 6 ***E.*** 5

24. A bar code is made of 16 black and white bars taking turns. It starts with a black bar and ends with a **white** bar. Each black bar is wide or narrow. There are 3 more white bars than wide black bars.
How many narrow black bars are there?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 5

Answer Key

1.	**D**	**2.**	**D**	**3.**	**C**	**4.**	**B**	**5.**	**A**	**6.**	**D**
7.	**E**	**8.**	**C**	**9.**	**A**	**10.**	**A**	**11.**	**A**	**12.**	**B**
13.	**C**	**14.**	**B**	**15.**	**C**	**16.**	**A**	**17.**	**A**	**18.**	**D**
19.	**B**	**20.**	**A**	**21.**	**C**	**22.**	**D**	**23.**	**B**	**24.**	**D**

Solutions

1. D (192) — Numbers with 1, 2, 3, ... digits: 9 × 1 + 90 × 2 + 1 × 3 = 192. (There are 90 two-digit numbers: 10 to 99.) [2007 Q23 · one_to_round]

2. D (20) — The border of an 6 × 6 square has 6 + 6 + 4 + 4 = 20 cells (or 6 × 6 – 4 × 4 = 20). [2007 Q21 · border_next]

3. C (1, 4 and 5) — At every fork Zita takes exactly one branch, so there must be one number from each fork: 1, 4 and 5 works. Each of the others has two numbers from the same fork. [2007 Q1 · which_group]

4. B (2 oranges) — Take 1 melon off both plates: 2 oranges balance 2 – 1 = 1 melon. So one melon = 2 oranges. [2002 Q9 · one_more_balances]

5. A — The side sewn to the pole always shows at its full length. In picture A part of the pole is bare, so a piece of the flag is missing. [2002 Q15 · torn_flag]

6. D (Leo) — The only possible matching: Sofia – piano; Noah – flute; Julien – violin; Leo – guitar. [2002 Q18 · who_has]

7. E (4) — Go backwards and do the opposite: 1 + 3 = 4, 4 × 3 = 12, 12 ÷ 3 = 4. [2007 Q5 · favourite_number]

8. C (4 m 52 cm) — 30 dm = 3 m and 50 mm = 5 cm, so the training jump was 3 m 35 cm. Adding 117 cm gives 4 m 52 cm. [2003 Q10 · jump]

9. A (12) — They sit on the edges, but not at the corners. 12 edges with 3 – 2 = 1 such cubes each: 12 × 1 = 12. [2007 Q17 · two_faces]

10. A (21) — The aquarium holds 3 × 3 × 3 = 27 cubes. There are already 6 inside (count column by column, also the hidden ones). 27 – 6 = 21. [2007 Q14 · how_many_more]

11. A (Lily) — Sara: 2 × 5 + 4 = 14; Martha: (8 – 6) × 7 = 14. Lily: with 1, 3 and 9 nothing gives 14 (the closest are 13 = 1 + 3 + 9 and 18 = (3 – 1) × 9). [2002 Q21 · who_cannot]

12. B — The cut corner is the middle of a side of the paper (look where the folds are). Mirrored four times it gives notches in the middle of the left and right sides: design B. [2007 Q24 · which_result]

13. C (6) — 2016 – 2009 = 7, but Leo's birthday (December 16) has not come yet that year, so Leo is 6. [2007 Q12 · age_on_day]

14. B (4) — The blue inside is 2 long and 2 tall, so one layer of it has 2 × 2 = 4 cubes. 8 ÷ 4 = 2 layers of blue, plus the red front and back: 4. [2003 Q9 · how_deep]

15. C — Only this piece is found in the line. The others don't fit — A: it has a pointed roof where the line has a round dome; B: it has two small bumps where the line has a pointed roof; D: it has a pointed roof where the line has two small bumps; E: it is a mirror image of a part of the line. [2002 Q7 · can_belong]

16. A (36 cm) — The two new edges (3 cm and 3 cm) replace the two removed pieces of the outer edge, which were also 3 cm and 3 cm. So the perimeter is the same as the rectangle's: 2 × (10 + 8) = 36 cm. [2002 Q6 · corner_cut]

17. A — All numbers add up to 36, so the pieces must be 11 + 12 + 13 = 36. Only this picture gives 11, 12, 13. [2002 Q11 · which_picture]

18. D (24) — There are 12 pages: 12 × 2 = 24 digits. [2003 Q16 · from_page]

19. B (1+2+5) — The empty area has 12 squares, so the three sizes must add up to 12. Only 1 + 2 + 5 also fit together in the right shape. [2003 Q18 · three_pieces]

20. A (8) — Tunnel: 16 × 4 = 64 cubes. Pyramid: 6 × 6 + 4 × 4 × 2 + 2 × 2 = 72. Missing: 72 – 64 = 8. [2002 Q20 · not_enough]

21. C — Two copies have 10 squares, so the rectangle is 1 × 10 or 2 × 5. Only two copies of piece C fit together like that. [2007 Q8 · two_copies]

22. D (12) — Think of Emma and Ben as one block: the block and the other 2 make 3 things to order, 3 × 2 × 1 = 6 ways. Inside the block Emma and Ben can swap, so 2 × 6 = 12. [2002 Q22 · side_by_side]

23. B (8) — List them by the first digit: 17, 26, 35, 44, 53, 62, 71, 80. That is 8. [2007 Q4 · count_with_sum]

24. D (3) — Starting black and ending white, the bars come in pairs: 8 black and 8 white. Wide: 8 – 3 = 5. Narrow: 8 – 5 = 3. [2003 Q17 · ends_differ]